

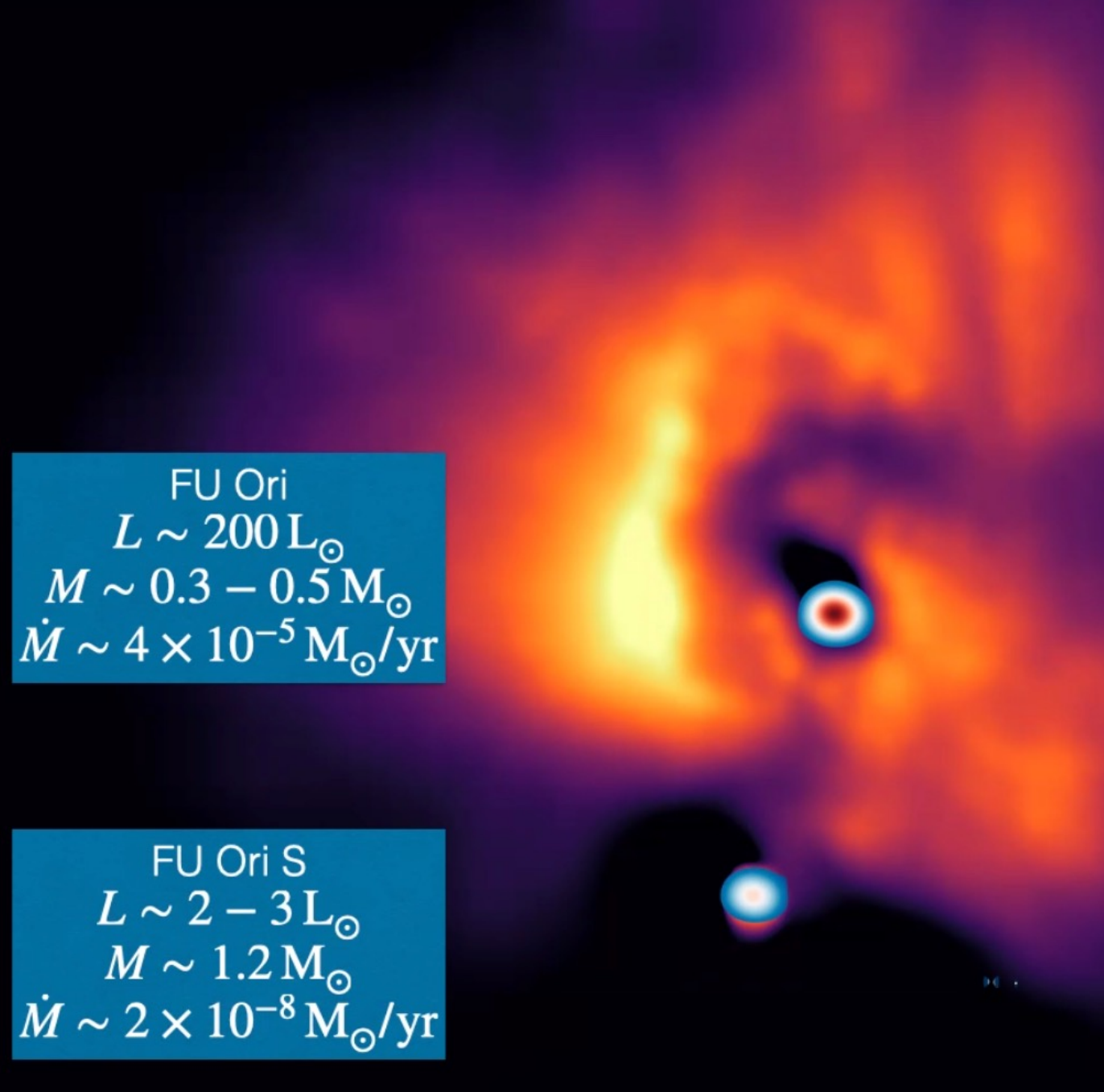


FU Ori Outbursts in growing binary systems

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Massive self-gravitating discs can fragment on numerous Jovian mass clumps, whose dynamics is very complex. VB10 showed that some can be accreted onto the central star, triggering FUor-type outbursts. In FU Ori itself, however, the famous outburst originates from the lower-mass secondary star. Our FARGO-ADSG simulations of self-gravitating discs show how this can happen naturally. We find that clumps are often accreted or disrupted by the secondary rather than the primary. We present one such example, tracking the clump's gas and planetesimal dynamics concurrently with secondary's star mass and orbital evolution.

1. Fragmenting disc simulation

We use FARGO-ADSG (Baruteau & Masset 08) to simulate fragmenting disc. The most massive clump grows by smooth accretion and mergers (Fig. 1), becoming the secondary star. We convert it into a sink particle (see step 3). At time 9.2 kyr, a clump collides it (Fig. 2). We assume that the clump made a solid core planet in the centre, which we inject in it. We find that the secondary has an FU Ori like outburst after the merger.

2. Evolution of the clumps

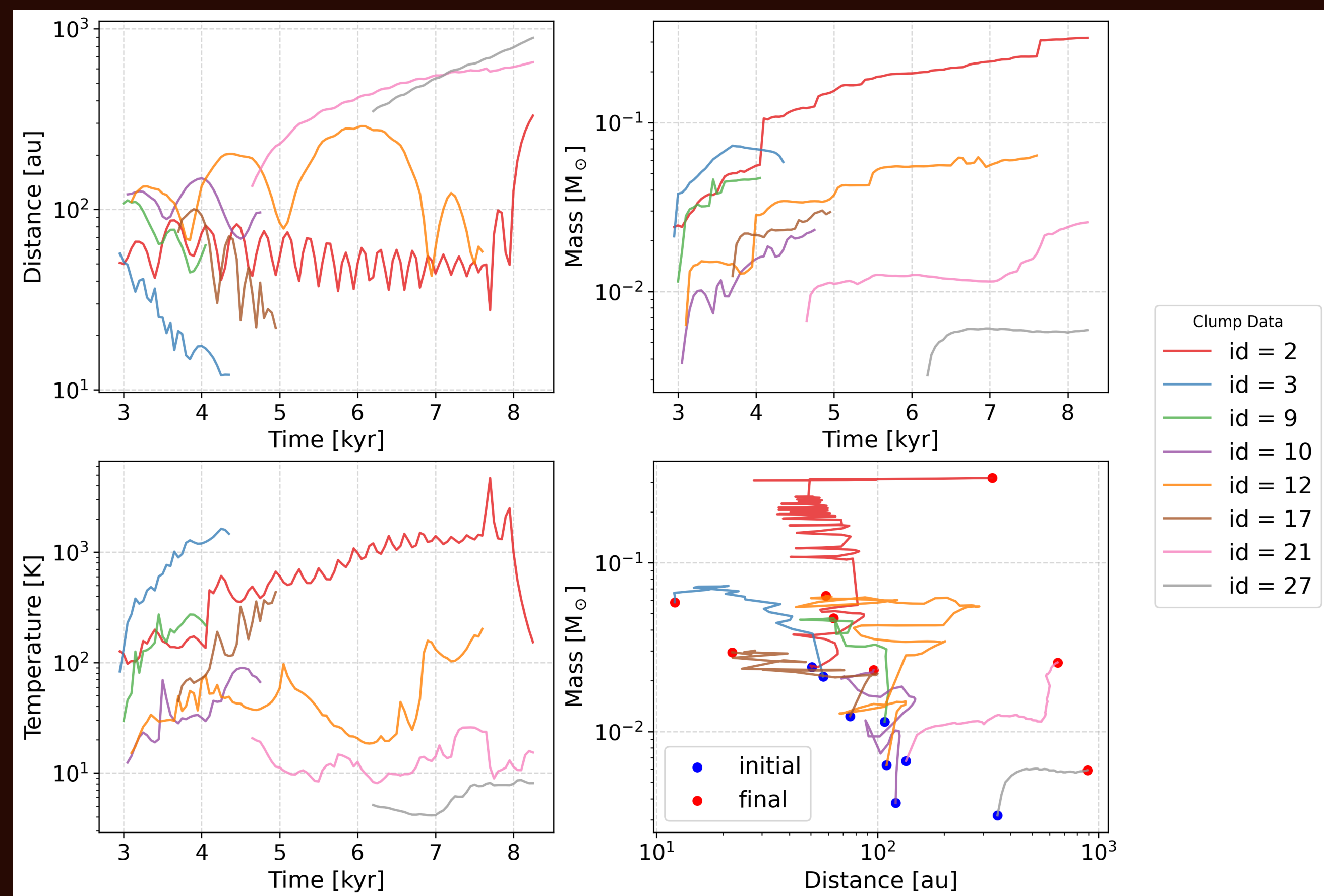


Fig. 1. Temporal evolution of (top-left) radial distance, (top-right) clump mass, (bottom-left) peak temperature, and (bottom-right) mass–distance of selected clumps. Blue dots mark clump formation or simulation start times; red dots indicate final states, including disruption or the simulation end.

3. The merger that fuels an outburst, an example

To model the secondary and the clump internal structure, we inject at 6 kyr:

- $1 M_J$ secondary with an accretion radius of 0.33, placed in a $0.2 M_{\odot}$ clump
- $1 M_{\oplus}$ planet (P1) is placed in the clump centre.

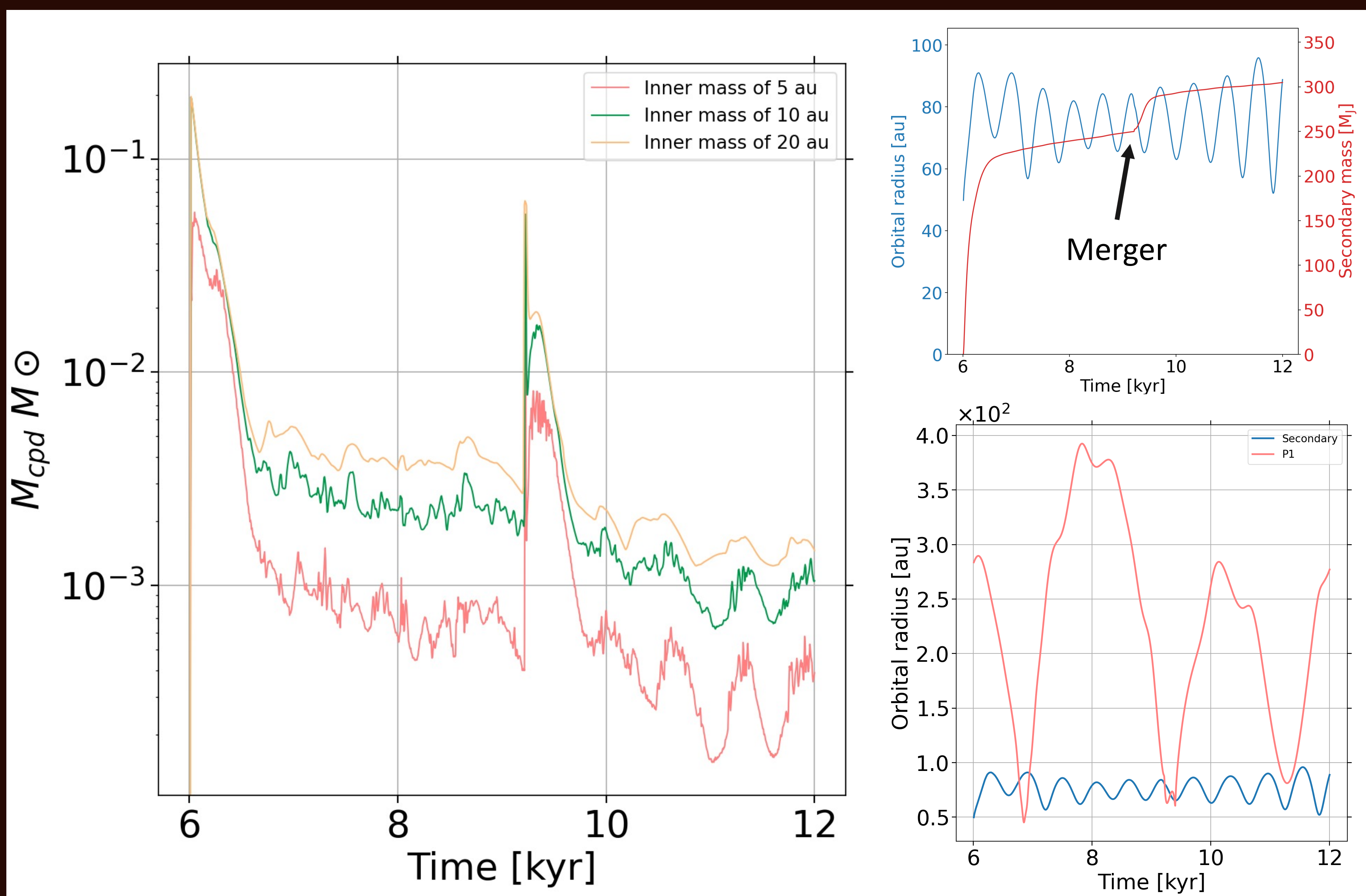
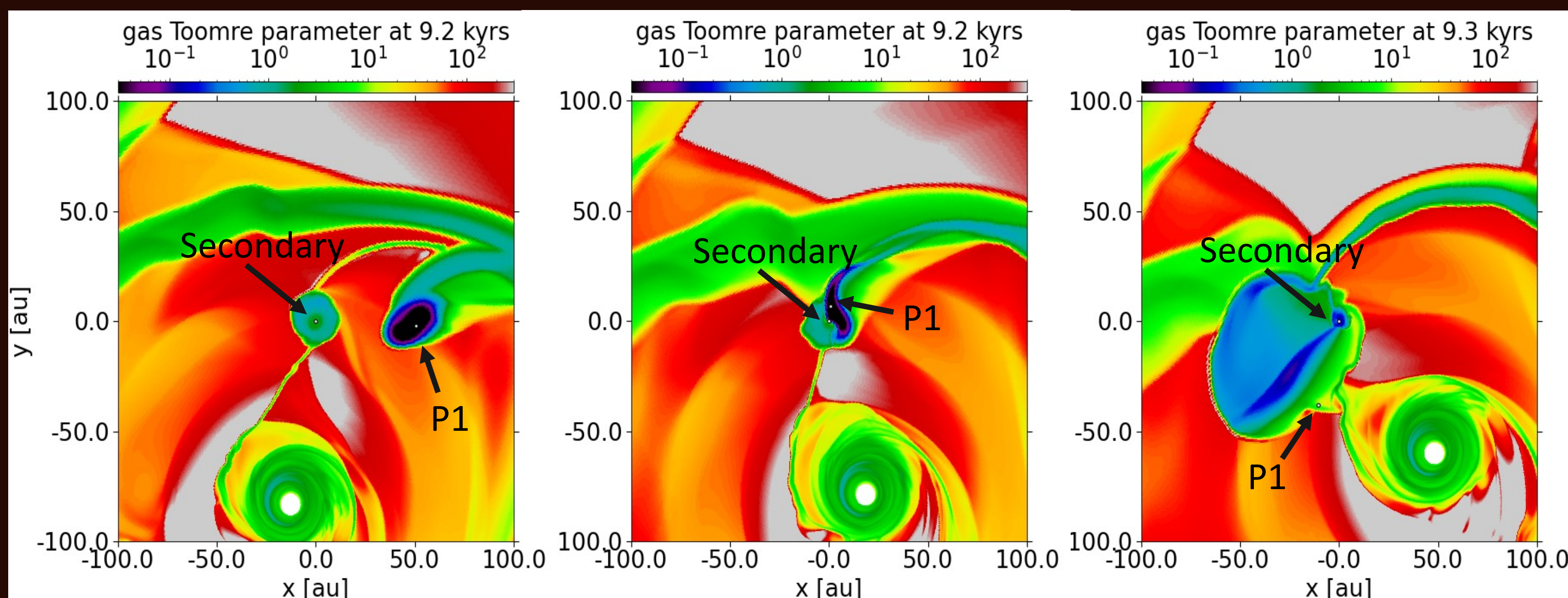


Fig 2. **Top:** The Toomre Q parameter of the disc just before, during, and after a clump merger with the secondary star. Darker regions are generally denser. **Bottom:** Time evolution of (left) circum-secondary star disc mass, (top-right) its orbital radius and mass, and (bottom-right) orbital radii of the secondary (blue) and the disrupted core of the clump

Conclusions:

- Clump–clump mergers may fuel secondary star growth and FUor-type outbursts on it.
- After mergers, some planetary cores are ejected as free-floating planets, some become wide-orbit planets.

References:

- [1] Vorobyov E. I., Basu S., 2010, ApJ, 719, 1896; [2] Baruteau, C., & Masset, F. 2008, ApJ, 678, 483